Date 10/08/2026

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Aim:- To write a LEX program to recognise identifiers, keywords & operators.

Algorithm

1. Start
2. Define keyword list
3. Define regex for identifier & operator
4. Read input using
5. Match patterns and print type.
6. Stop

Program Theory:- In programming languages, identifiers are names given to variable & functions. Keywords are reserved word like int, float etc. Operators are symbols used to perform operation. LEX helps distinguish between these categories using patterns matching, keywords are defined explicitly. Identifiers follow naming rules. Operators are recognised using symbols.

Program

```
%{
#include <stdio.h>
%}
%%
"int "|"float"|"if"|"else"
    {
      printf("keywords : %s\n", yytext);
    }
```

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Output

Input : int a = b + c

Keyword : int
Identifier : a
Operator : =
Identifier : b
Operator : +
Identifier : c

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```
[a-zA-Z_][a-zA-Z0-9_]*
{
printf("Identifier : %s\n", yytext);
}
[+|-|*|/|=]
{
printf("Operator : %s\n", yytext);
}
[\t\n]+
{
}
{
}
%%

int main()
{
yylex();
return 0;
}
```

Viva Question

Q1 How do you differentiate between keywords & identifiers in LEX?

Ans Keyword are reserved words of a programming language, such a int, if, while. Identifiers are names given to variables, function etc. such as a, sum, or count. In LEX, seperate pattern/rules can be given to recognize keyword & identifiers.

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Q2 What regular expression do you use to recognize identifiers?

Ans A common regular expression for identifiers is [a-zA-Z_][a-zA-Z0-9_]*.
It means the identifiers must start with a letter or underscore. After that it can contain letters, digits & underscores.

Q3 How do you handle C operators like ++ & == ?

Ans C operators can be recognized by writing their pattern explicitly in LEX rules. For example, \+\+ recognizes the increment operator ++, while == recognizes equality. The longer operator should be matched before its shorter form when necessary.

Q4 What is the difference between a token & a lexeme?

Ans A token is the category or type of a meaningful unit, such as KEYWORD or IDENTIFIER. A lexeme is the actual sequence of characters matched in the input.
For example, in int a, int is a lexeme belonging to the 'keyword' token.

Q5 How do you handle escape sequence in string literal

Ans String literals can be recognized using a regular expression the allowed normal characters & escape sequences. for ex, \"([^\"\\]|\\.)*\" can recognize a quoted string containing escaped characters. Escape sequence such as

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\n, \t, \" handled as part of string pattern.